An experimental comparison of at-issueness diagnostics

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Abstract At-issueness is a key concept in semantics and pragmatics, used to distinguish the main point of an utterance from background information. However, there is no consensus on how at-issueness should be defined, and different diagnostics reflect different assumptions about its underlying nature. This paper presents a systematic experimental comparison of five diagnostics of at-issueness, applied to the same target contents, including appositive non-restrictive relative clauses and the complements of clause-embedding predicates. The results reveal substantial differences between diagnostics, indicating that they are not interchangeable. In particular, diagnostics differ in whether they detect at-issueness differences between particular expressions, and that diagnostics measuring the at-issueness of contents embedded in questions yield greater by-content differentiation than diagnostics embedding them in assertions. This generalization suggests that the speech act used to present a content plays a key role in affecting diagnostic outcomes. We argue that this effect of speech act on at-issueness ratings is due to how questions and assertions relate at-issue and not-at-issue content to speaker commitments, and cannot be explained by different notions of at-issueness assumed in the literature.

Keywords: at-issueness, diagnostics, experimental pragmatics

1 Introduction

At-issueness is a key concept in theoretical semantics and pragmatics,[1] used in the analysis of various phenomena including presupposition, conventional implicature, evidentials, expressives, NPI licensing, and co-speech gestures (e.g., Faller 2002; 2019; Horn 2002; Potts 2005; Simons et al. 2010; Lee 2011; Tonhauser et al. 2018; Esipova 2019; Korotkova 2020; Barnes & Ebert 2023). It is generally understood as distinguishing the main point of an utterance (at-issue content) from propositions conveying background information (not-at-issue content). However, there is no consensus on how at-issueness should be defined, and different diagnostics reflect different assumptions about its underlying nature (e.g., Tonhauser 2012; Snider 2017a; b; 2018; Tonhauser et al. 2018; Koev 2018; Faller 2019; Esipova 2019; Korotkova 2020). Consequently, empirical claims about whether a given expression contributes at-issue or not-at-issue content may be relative to specific diagnostics, and different diagnostics may not target the same underlying phenomenon.

1.1 Definitions of at-issueness

One influential definition assumes that discourse is structured around addressing a *Question Under Discussion* (QUD) (Roberts 2012; Ginzburg 1996) and treats at-issue content as the part of an utterance intended to address that question (Amaral et al. 2007; Simons et al. 2010):

(1) QUD at-issueness: (based on Simons et al. 2010, p. 323)
A content *m* is at-issue in a context *c* iff
a. *m* is relevant[2] to the QUD in *c*, and
b. the speaker has a recognizable intention to address the QUD via *m*.

[1] Potts (2015) credits the term to Bill Ladusaw, "who began using it informally in his UCSC undergraduate classes in 1985" (p. 2).

Another definition, given in (2), is grounded in understanding assertion as a proposal to update the common ground (Stalnaker 1978; Clark & Schaefer 1989; Ginzburg 1995; Farkas & Bruce 2010). Here, the at-issue content of an assertion is proposed for addition to the common ground (Koev 2013), while not-at-issue content is presupposed (already entailed in the common ground; Stalnaker 1973; 2002), or added without being proposed (Murray 2010; 2014; AnderBois et al. 2010; 2015).

(2) Proposal at-issueness: (Koev 2013, pp. 50–51)
A proposition p is at-issue in a context c iff
a. p is a proposal in c and
b. p has not been accepted or rejected in c.

These definitions differ in their theoretical grounding and there is no consensus on whether they characterize distinct underlying notions of at-issueness or perspectives on a single phenomenon.[3] Some proposals suggest a unified view: if proposing content for the common ground is intrinsically linked to managing the QUD (e.g., Farkas & Bruce 2010), the proposal-based definition can be seen as a special case of the QUD-based definition, applied to assertions (AnderBois et al. 2015; Faller 2019). Others have argued that the definitions may correspond to distinct underlying phenomena, motivated by empirical differences between at-issueness diagnostics (Koev 2018).

1.2 At-issueness diagnostics

Diagnostics of at-issueness operationalize the theoretical notions in (1) and (2) via linking assumptions about how at-issue content behaves in discourse. These can be organized around three properties of at-issue content identified in Tonhauser (2012) as motivating diagnostic strategies:

(i) at-issue content addresses the question under discussion (see also Amaral et al. 2007; Simons et al. 2010),
(ii) the at-issue content of questions determines the relevant set of alternatives, and
(iii) at-issue content can be directly assented or dissented with (see also Shanon 1976; Groenendijk & Stokhof 1984; Faller 2002; 2006; Murray 2010).

Four commonly used diagnostics are illustrated in (3–6) for sentence-medial appositive nonrestrictive relative clauses (NRRCs), which are typically taken to contribute not-at-issue content (Potts 2005).

(3) QUD diagnostic
Nora: *What did Greg buy?*
Leo: *Greg, who bought a new car, is envied by his neighbor.*
Question to participants: How well does Leo's response fit Nora's question?

(4) 'asking whether' diagnostic
Nora: *Is Greg, who bought a new car, envied by his neighbor?*
Question to participants: Is Nora asking whether Greg bought a new car?

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[2] m may be a proposition or a question denotation, and relevance to the QUD can be defined in terms of contextual entailment of a partial or complete answer (Roberts 2012; Simons et al. 2010), or based on inferences about the probability of QUD-answers (Büring 2003; Beaver & Clark 2008).

[3] Further definitions proposed in the literature include those based on restrictive vs. non-restrictive modification (Esipova 2019; 2021), and coherence-based discourse structure (Hunter & Asher 2016; Jasinskaja 2016). Clarifying the relationships among the various proposed definitions of at-issueness remains an important question for future research.

(5) Direct-dissent diagnostic

Nora: *Greg, who bought a new car, is envied by his neighbor.*
Leo: *No, that's not true, he didn't buy a new car.*

Question to participants: How natural is Leo's rejection of Nora's utterance?

(6) 'yes, but' diagnostic

Nora: *Greg, who bought a new car, is envied by his neighbor.*
Leo: *Yes, but he didn't buy a new car.* /
Yes, and he didn't buy a new car. /
No, he didn't buy a new car.

Task for participants: Choose the response that sounds best.

These diagnostics can be understood as targeting properties (i)–(iii). The QUD-diagnostic (3) (Amaral et al. 2007; Lee 2011; Tonhauser 2012; Chen 2024) targets property (i) by testing whether the target content can felicitously address an established QUD. It presents the content in a response to a question that makes the the target content relevant and elicits ratings of question-answer fit. Ratings are expected to be high when the content can be construed as at-issue and low otherwise. Accordingly, ratings are expected to be low for the appositive NRRC in (3).

The 'asking whether' diagnostic (4) (Tonhauser et al. 2018; Solstad & Bott 2024; Degen & Tonhauser 2025) targets property (ii) by probing whether the target content determines the alternatives of a polar question. It presents the target content in a polar question, and elicits judgments about whether the speaker is asking whether that content is true. These judgments are expected to be high for at-issue content and low otherwise. Accordingly, participants are expected to judge that the speaker is not asking about the NRRC content in (4).

The direct-dissent diagnostic (5) (Faller 2002; 2006; Papafragou 2006; Amaral et al. 2007; Murray 2010; 2014; AnderBois et al. 2010; 2015; Tonhauser 2012; Syrett & Koev 2015), and the 'yes, but' diagnostic (6) (Xue & Onea 2011; Cummins et al. 2013; Destruel et al. 2015) target property (iii). Both present the target content in a declarative sentence and assume that only at-issue content can be directly assented or dissented with, whereas dissenting with not-at-issue content requires more indirect discourse moves. The direct-dissent diagnostic elicits naturalness ratings for a response directly dissenting with the target content, which are expected to be high for at-issue content and low otherwise. Accordingly, low ratings are expected for the NRRC content in (5). The 'yes, but' diagnostic presents a forced choice between *'no'*, *'yes, but'*, and *'yes, and'* when denying the target content. Under the assumption that the *'yes'*-responses provide indirect denials suitable for dissenting with not-at-issue content (Xue & Onea 2011), participants are expected to prefer one of the *'yes'*-responses when denying not-at-issue content, such as the NRRC content in (6).[4]

1.3 Diagnostic differences and notions of at-issueness

The question whether different definitions of at-issueness characterize a single phenomenon or distinct ones is also reflected in competing views about whether diagnostics probe distinct underlying notions or provide different ways of probing a single, unified phenomenon.

[4] Predictions for at-issue content are less clear and may be context-dependent. Although at-issue content permits direct dissent, it does not require it, and indirect dissent is also possible (see, e.g., Cummins et al. 2013 on Shanon's 1976 "Hey, wait a minute" diagnostic). Work in conversation analysis suggests that preferences for direct versus indirect dissent vary with context and content. Direct dissent is often overall dispreferred (Sacks 1987; Brown & Levinson 1978; Pomerantz 1984; Locher 2004), but the strength of this preference can vary with cultural norms (e.g., Kakava 2002) or speaker gender (e.g., Tannen 1990). In contrast, direct dissent can be preferred in adversarial legal contexts (Atkinson & Drew 1979), in disputes (Kotthoff 1993; Kangasharju 2009), or in responses to self-deprecating statements (Pomerantz 1984).

Tonhauser (2012) suggests that diagnostics targeting properties (i)–(iii) can be taken to probe a common underlying notion of at-issueness, based on the QUD-based definition.

In contrast, views that take different diagnostics to probe distinct phenomena are motivated by empirical divergences between them. For example, although appositive NRRCs are widely taken to contribute not-at-issue content, a forced-choice continuation study by Syrett & Koev (2015) found that sentence-final appositive NRRCs pattern as more at-issue than medial ones, under a version of the direct-dissent diagnostic, which targets property (iii). Snider (2017b) argues that diagnostics targeting property (iii) are sensitive to this medial-final distinction, but diagnostics targeting properties (i) or (ii) are not. Similar diagnostic differences have been reported for sentence-final slifting parentheticals (Koev 2018) and evidential inferences (Korotkova 2020).

Such differences have been interpreted in two main ways. Koev (2018) takes them to indicate that at-issueness is not a uniform notion. On this view, the QUD-diagnostic (property i) targets a QUD-based notion of at-issueness, while the direct-dissent and 'yes but' diagnostics (property iii) probe a proposal-based notion. The two definitions are thus taken to characterize distinct discourse phenomena, which may diverge empirically. Another view (Snider 2017b; a; 2018; Korotkova 2020) maintains a unified notion of at-issueness, arguing that diagnostics targeting properties (i) and (ii) genuinely probe at-issueness, whereas property-(iii) are instead sensitive to the availability of the target content for propositional anaphora. Property-(iii) diagnostics are thus taken to reflect whether the target content can provide antecedents for response particles (*yes/no*) or propositional pro-forms (e.g., *that* in *that's not true*), which are present in the direct-dissent and 'yes but' diagnostics.

Although empirical differences between at-issueness diagnostics have been reported across a range of constructions, they have not, to date, been evaluated in a systematic experimental comparison that applies multiple diagnostics to the same contents. Moreover, there is currently no agreed-upon definition of at-issueness, and existing diagnostics may or may not be probing the same underlying phenomenon. This raises two central questions:

i. Do different diagnostics yield consistent results when applied to the same contents?
ii. Do different diagnostics reflect distinct underlying notions of at-issueness?

1.4 Previous findings

Initial experimental work on these questions has yielded mixed answers. As noted above, Syrett & Koev (2015) found a positional effect for appositives under a version of the direct-dissent diagnostic. Participants saw declarative sentences introducing the target content and chose between two *no*-responses expressing direct dissent, targeting the appositive or the main-clause content (7).

(7) Syrett & Koev 2015, p. 17

A: *My friend Sophie, a classical violinist, performed a piece by Mozart.*
B1: *No, she's not.* (target: NRRC)
B2: *No, she didn't.* (target: main clause)

Direct dissent targeting the at-issue main clause (B2) is assumed to be always available. When the target content is not-at-issue, as with the NRRC content in (7), the B1 response is expected to be strongly dispreferred. Indeed, participants strongly preferred B2 responses (with proportions around 79%) when the competing alternative targeted a sentence-medial NRRC. Crucially, this preference was weaker for sentence-final NRRCs (around 65%), suggesting that final NRRCs may be interpreted as more at-issue than medial ones. Since this positional contrast for NRRCs has been argued to be specific to assent and dissent-based (property iii) diagnostics, a key question is whether it also emerges with other diagnostics. Therefore, one concrete way in which we address our research question i., whether different diagnostics yield consistent results, is testing whether they reproduce the positional effect for appositive NRRCs observed in Syrett & Koev 2015.

Beyond appositives, Tonhauser et al. (2018) compared the 'asking whether' and 'are you sure?' diagnostics across several English expressions, including sentence-medial appositives and complements of the clause-embedding predicates *annoyed, discover* and *know* (8).

(8) Tonhauser et al. (2018), p. 503

a. annoyed: *Martha's neighbor is annoyed that Martha has a new BMW.*
b. discover: *Mary discovered that her daughter has been biting her nails.*
c. know: *Billy knows that Martha has a new BMW.*

The 'are you sure?' diagnostic presents participants with dialogues like (9) and collects ratings about whether the response addresses the question (c.f. the 'really' test of Shanon 1976).

(9) Tonhauser et al. (2018), p. 519
Fred: *Martha's new car, a BMW, was expensive.*
Carla: *Are you sure?*
Fred: *Yes, I am sure that Martha's new car is a BMW.*

This diagnostic is taken to target property (iii), by testing whether the target content can be directly challenged. The target content is introduced by the initial assertion, the question challenges this assertion, and the response reinforces the target content. When the target content is at-issue, this response is expected to be judged as addressing the question, and as not addressing the question otherwise. The diagnostic is therefore expected to yield low ratings for the NRRC in (9).

The authors found both convergence and divergence in their comparison (p. 526ff.). Ratings were correlated across diagnostics, suggesting that they may be sensitive to a shared underlying phenomenon. However, the 'asking whether' diagnostic yielded lower overall ratings and smaller differentiation between contents than the 'are you sure?' diagnostic.

Subsequent work shows that the 'asking whether' diagnostic can nonetheless detect fine-grained differences. Degen & Tonhauser (2025) found fine-grained at-issuness differences among the contents of the complements of 20 clause-embedding predicates in English. Figure 1 shows the mean 'asking whether' ratings for the 20 predicates used in their study, ordered by mean rating.

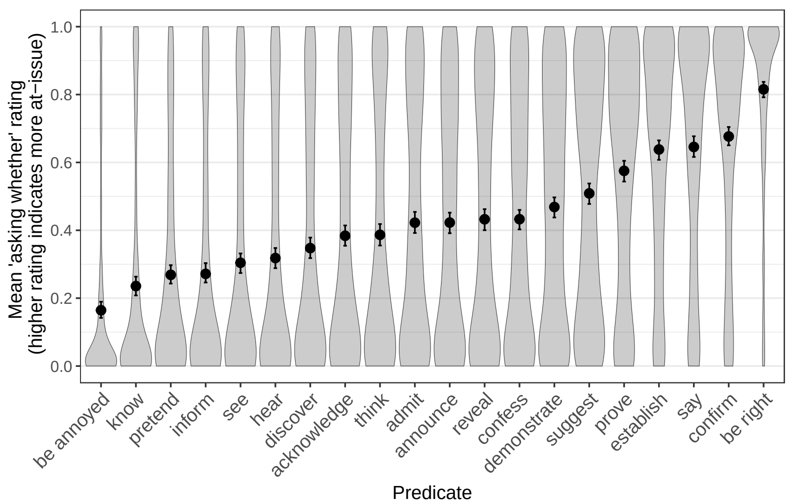


Figure 1: Mean 'asking whether' ratings for the contents of the clausal complements of 20 clause-embedding predicates, from Degen & Tonhauser 2025. Error bars indicate 95% bootstrapped confidence intervals. Violin plots indicate the distribution of individual ratings.

Here, the complement of *annoyed* was rated as less at-issue than those of the other predicates tested, whereas that of *be right* received the highest ratings, and there were fine-grained by-predicate differences between many of the predicates in between.

Because the previous research indicates that diagnostics may differ in how strongly they differentiate among contents, a second way in which we concretely address our research question i., about whether different diagnostics yield consistent results, is testing whether they reproduce fine-grained by-content differences for clause-embedding predicates observed in Degen & Tonhauser 2025.

1.5 Outlook

This paper presents a systematic experimental comparison of at-issueness diagnostics across six experiments, taking a first step towards addressing the two overarching questions repeated here:

i. Do different diagnostics yield consistent results when applied to the same contents?
ii. Do different diagnostics reflect distinct underlying notions of at-issueness?

To address (i), we compare diagnostics by testing whether they reproduce by-content contrasts reported in prior work, namely the medial-final contrast for appositive NRRCs (Syrett & Koev 2015) and fine-grained differences among clause-embedding predicates (Degen & Tonhauser 2025). To begin addressing (ii), we consider how any diagnostic differences should be explained. While convergence would support a single-notion view, divergence would raise the question whether the differences align with proposed distinctions, such as QUD-based vs. proposal-based at-issueness or at-issueness vs. anaphoric availability, or whether they require an independent explanation.

In what follows, Section 2 reports on Experiments 1–4, which apply the four diagnostics in (3)–(6) to seven contents in English, including medial and final appositive NRRCs and the complements of selected clause-embedding predicates from Degen & Tonhauser (2025) (*confess, confirm, discover, know, be right*). None of these four experiments reproduced the positional effect for NRRCs reported by Syrett & Koev (2015). Only Exp. 2 ('asking whether') came close to observing the by-predicate differences found in Degen & Tonhauser (2025), with differences between *know, discover, confess, confirm, be right* and increasing mean at-issueness ratings in that order. The results suggest that diagnostics do vary in whether they detect at-issueness differences between particular expressions. Notably, the 'asking whether' diagnostic, which showed the greatest by-content differentiation in Exp. 2, is the only diagnostic which directly elicits judgments of speaker intentions, and the only one where the target content is embedded in a polar question.

Section 3 asks whether the the greater differentiation under the 'asking whether' diagnostic was due to the polar question embedding or the response task. Experiments 5 and 6 compared the 'asking whether' diagnostic with a direct-response diagnostic (Amaral et al. 2007; Tonhauser 2012) that also embeds content in a polar question but uses a different response task, eliciting acceptability judgments. Using the 20 clause-embedding predicates from Degen & Tonhauser (2025), the two experiments yielded highly correlated results, indicating that the observed high differentiation in Exp. 2 was driven by the polar question embedding, not the response task.

The clearest difference we found is therefore between diagnostics that embed the target content in polar questions and those that embed it in declaratives. Diagnostics with polar question embedding, like the 'asking whether' diagnostic (Exps. 2, 5) and the direct-response diagnostic in (Exp. 6), show greater differentiation between contents than diagnostics with declarative embeddings, like the direct-dissent and 'yes, but' diagnostics (Exps. 3–4). This pattern is not predicted by a dual-notion view of at-issueness (QUD-based vs. proposal-based) or by the distinction between at-issueness and anaphoric availability. Instead, we argue that the speech act used to present the target content plays a central role: we propose that the illocutionary force of declarative sentences may interact with at-issueness ratings in a way that decreases by-content differentiation, while the illocutionary force of polar questions does impact the detection of differences between contents. Section 4 provides further discussion of this point, and Section 5 concludes.

2 Experiments 1-4

To compare the results of at-issueness diagnostics,[5] we conducted four experiments that each measured at-issueness with a different diagnostic, namely the QUD diagnostic (Exp. 1), the 'asking whether' diagnostic (Exp. 2), the direct-dissent diagnostic (Exp. 3) and the 'yes, but' diagnostic (Exp. 4).[6] Each experiment tested the same set of contents associated with the seven types of expressions in (10): sentence-medial and sentence-final NRRCs (10a–b), as well as the contents of the clausal complements of *be right, discover, confess, confirm, discover,* and *know* (10c–g).

(10) a. Content of sentence-medial NRRC
Lucy, who broke the plate, apologized. ⇝ Lucy broke the plate

b. Content of sentence-final NRRC
The police found Jack, who saw the murder. ⇝ Jack saw the murder

c. Content of the clausal complement of *know*
Ann knows that Raul cheated on his wife. ⇝ Raul cheated on his wife

d. Content of the clausal complement of *discover*
Mary discovered that Denny ate the last cupcake. ⇝ Denny ate the last cupcake

e. Content of the clausal complement of *be right*
Tom is right that Ann stole the money. ⇝ Ann stole the money

f. Content of the clausal complement of *confirm*
Harry confirmed that Greg bought a new car. ⇝ Greg bought a new car

g. Content of the clausal complement of *confess*
Lucy confessed that Dustin lost his key. ⇝ Dustin lost his keys

These seven contents were chosen because prior literature observed differences in at-issueness between two or more of these contents using a particular diagnostic for at-issueness. As discussed in §1, Syrett & Koev 2015 observed differences between sentence-medial and -final NRRCs using a variant of the direct-dissent diagnostic, and Degen & Tonhauser 2025 observed differences between *be right, discover, confess, confirm, discover,* and *know* using the 'asking whether' diagnostic. Thus, comparing these seven contents across the four diagnostics in Exps. 1–4 will allow us to assess whether the differences that emerge from one diagnostic also emerge from others. In each experiment, participants read the stimuli and gave ratings corresponding to the diagnostics.

Following prior work (e.g., Tonhauser et al. 2018), we describe higher/lower mean ratings by saying that content is more/less at-issue under a given diagnostic. However, we remain agnostic about whether at-issueness is an underlyingly binary or a gradient property (cf. Tonhauser et al. 2018; Barnes & Ebert 2023). On a gradient view, gradient mean ratings may be taken to reflect gradient at-issueness differences. On a categorical view, gradient mean ratings could be taken to reflect uncertainty about what the relevant QUD or proposal is, thus indicating the frequency or ease with which a particular QUD is attributed to the utterance.

2.1 Methods

2.1.1 Participants

For each of the four experiments, we recruited 80 unique participants on Prolific. These participants had registered on the platform as living in the USA and as having English as their primary language.

[5] We use the term "at-issueness diagnostic" descriptively throughout the paper, even though our findings may ultimately suggest that some diagnostic tracks a different theoretical construct. We return to this issue in the general discussion.

[6] The experiments, data and R code for generating the figures and analyses of the experiments reported in this paper are available at https://anonymous.4open.science/r/at-issueness-diagnostics-232C/.

They had at least 50 previous submissions and an approval rate of at least 97%. Table 1 shows the age and gender distributions of the recruited participants.

	recruited	ages (mean age)	f/m/nb/dnd
Exp. 1 (QUD)	80	18-81 (43.8)	42/37/0/1
Exp. 2 (asking whether)	80	20-74 (38.5)	48/30/1/1
Exp. 3 (direct dissent)	80	18-77 (39.1)	50/28/1/1
Exp. 4 (yes, but)	80	19-67 (38.0)	48/30/2/0

Table 1: Information about the participants recruited in Exps. 1-4 (f = female, m = male, nb = nonbinary, dnd = did not disclose).

2.1.2 Materials and procedure

As shown in Fig. 2, the response options differed by diagnostic. In Exp. 1 (QUD diagnostic, panel (a)), participants rated how well the response fit the question on a slider marked 'totally doesn't fit' on one end (coded 0) and 'totally fits' on the other end (coded as 1). In Exp. 2 ('asking whether' diagnostic, panel (b)), participants judged whether the question was about the target content, using a slider marked 'no' on one end (coded as 0) and 'yes' on the other (coded as 1). In Exp. 3 (direct-dissent diagnostic, panel (c)), participants rated the naturalness of the direct dissent on a slider marked 'totally unnatural' (coded as 0) on one end and 'totally natural' on the other (coded as 1). Finally, in Exp. 4 ('yes, but' diagnostic, panel (d)), participants chose the response that sounded best; the two indirect dissents were coded as 0 and the direct one as 1. Across the four experiments, the responses were coded so that 1 meant that the content to be diagnosed was rated as at-issue and 0 as not-at-issue.

Each of the seven contents in (10) was realized as one of the seven items in (11) in each of the four experiments.

(11)
a. Jack saw the murder.
b. Raul cheated on his wife.
c. Ann stole the money.
d. Danny ate the last cupcake.
e. Lucy broke the plate.
f. Dustin lost his key.
g. Greg bought a new car.

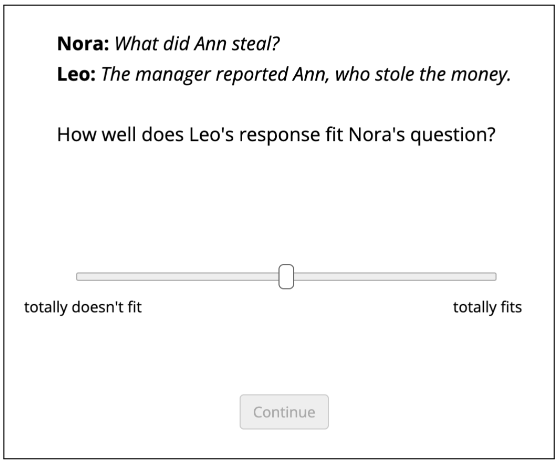
Each experiment included two control stimuli serving as attention checks: one was expected to receive a response at one end of the slider (Exps. 1-3) or a 'no' response (Exp. 4); the other control stimulus was expected to receive a response at the other end of the slider (Exps. 1-3) or a 'yes' response (Exp. 4). See Supplement A for the control stimuli used in Exps. 1-4.

In all four experiments, each participant's set of items was generated by randomly realizing each of the seven contents in (10) using a unique item in (11). Participants completed a total of 9 trials, namely 7 target trials and the same 2 control trials. Trial order was randomized for each participant.

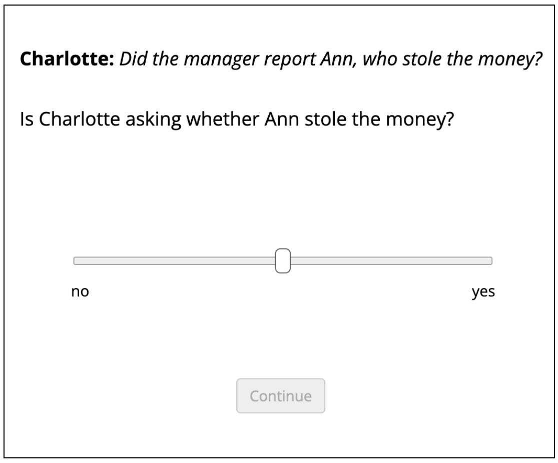
After completing the experiment, participants filled out a short optional demographic survey. To encourage truthful responses, participants were told that they would be paid no matter what answers they gave in the survey.

2.1.3 Data exclusion

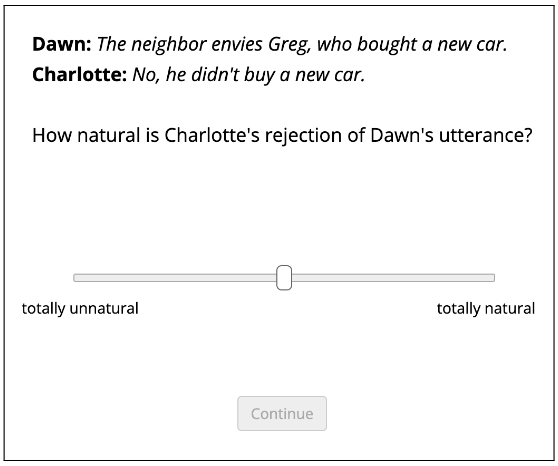
We excluded the data of participants that were not self-reported native speakers of American English and of participants whose responses to one of the two control trials was more than 2 sd away from the group mean (Exps. 1–3) or whose responses to one of the two control trials was incorrect (Exp. 4). Table 2 shows how many participants were excluded in each experiment, demographic



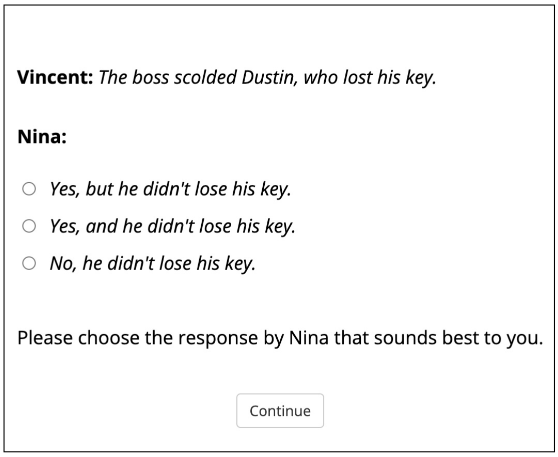
(a) Exp. 1: QUD diagnostic



(b) Exp. 2: 'asking whether' diagnostic



(c) Exp. 3: direct-dissent diagnostic



(d) Exp. 4: 'yes, but' diagnostic

Figure 2: Sample trials in (a) Exp. 1, (b) Exp. 2, (c) Exp. 3, and (d) Exp. 4.

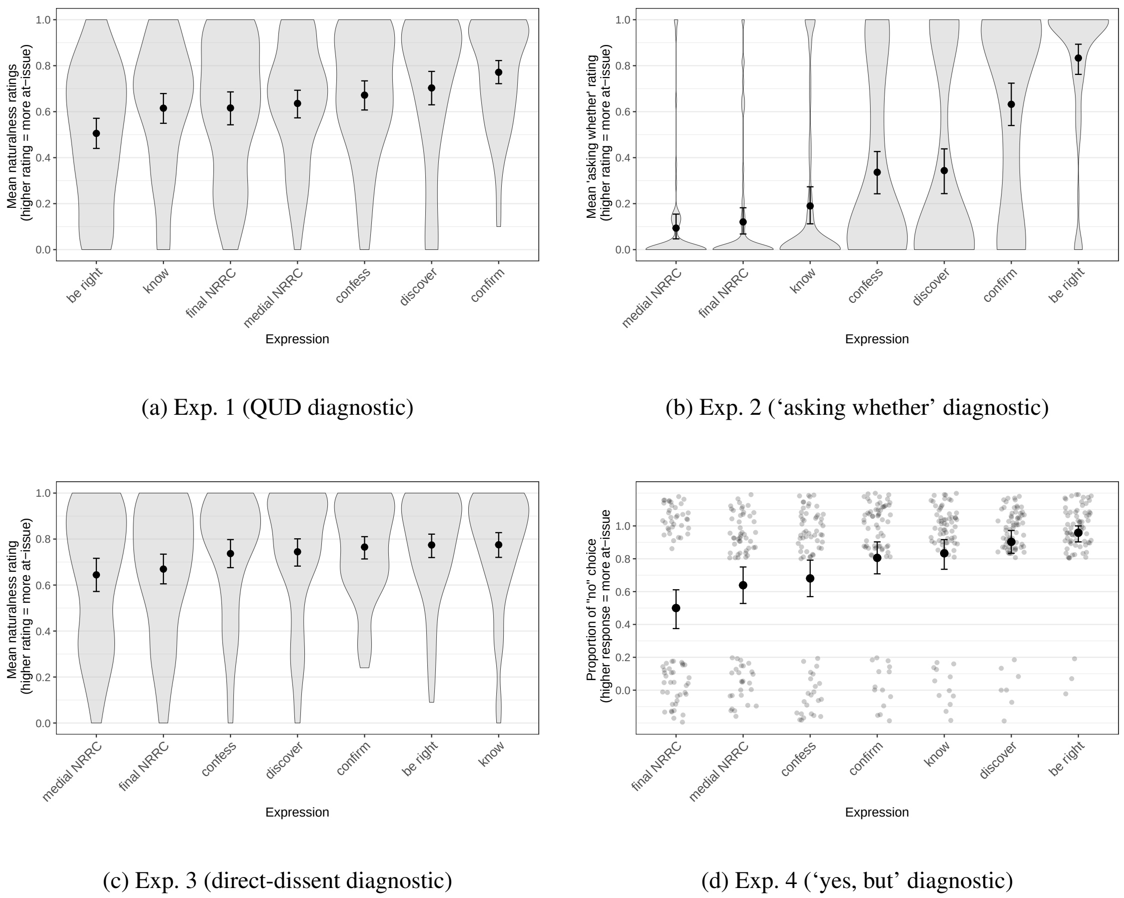
information for the remaining participants, and the number of data points that entered into the analyses.

	exclusion criterion		**remaining participants**		data
	language	fillers	ages (mean age)	f/m/nb/dnd	points
Exp. 1 (QUD)	1	10	18-81 (41.1)	36/32/0/1	621
Exp. 2 (asking whether)	2	4	22-74 (38.7)	45/27/1/1	666
Exp. 3 (direct dissent)	2	7	18-77 (39.5)	44/25/1/1	639
Exp. 4 (yes, but)	4	4	19-67 (38.5)	43/27/2/0	648

Table 2: Information from Exps. 1-4 about the number of participants whose data was excluded based on their self-declared language (variety) and the fillers, about the remaining participants, and about the number of data points that entered into the analysis.

2.2 Results

Fig. 3 plots the results of the four experiments by the expression that is associated with the seven target contents. Panel (a) shows the mean naturalness ratings in Exp. 1 (QUD diagnostic), panel (b) mean ‘asking whether’ ratings in Exp. 2 (‘asking whether’ diagnostic), panel (c) shows mean naturalness ratings in Exp. 3 (direct-dissent diagnostic) and panel (d) the proportion of ‘no’ choices in Exp. 4 (‘yes, but’ diagnostic).



(a) Exp. 1 (QUD diagnostic)

(b) Exp. 2 (‘asking whether’ diagnostic)

(c) Exp. 3 (direct-dissent diagnostic)

(d) Exp. 4 (‘yes, but’ diagnostic)

Figure 3: Results of Exps. 1–4. Panels (a)–(c) show the mean responses by expression for (a) Exp. 1 (QUD diagnostic), (b) Exp. 2 (‘asking whether’ diagnostic), and (c) Exp. 3 (direct-dissent diagnostic); panel (d) shows the proportion of ‘no’ choices by expression in Exp. 4 (‘yes, but’ diagnostic). Error bars indicate 95% bootstrapped confidence intervals. Violin plots in panels (a)-(c) show the kernel probability density of individual participants’ ratings. Gray dots in panel (d) represent individual participant responses (‘no’ vs. ‘yes’, jittered vertically and horizontally for legibility).

As is visually apparent in Fig. 3, the diagnostics differ in the overall range of mean responses across the seven contents (given as the difference between the largest and smallest by-content means). The range is largest in Exp. 2 (‘asking whether’ diagnostic), at .74 (from .1 to .83) and smallest in Exp. 3 (direct-dissent diagnostic), at .13 (.64 to .78). The results of Exp. 1 (QUD diagnostic), with a range of .27 (.51 to .77) and Exp. 4 (‘yes, but’ diagnostic), with a range of .46 (.5 to .96), fall in between.

The experiments differ in whether they reproduce distinctions between contents reported in prior work. Syrett & Koev (2015), using a variant of the direct-dissent diagnostic, found that sentence-medial NRRCs are more not-at-issue than sentence-final ones. In contrast, as shown in Fig. 3, no such difference is observed in Exps. 1–3; and in Exp. 4 ('yes, but' diagnostic), we find the opposite: sentence-final NRRCs are rated less at-issue than sentence-medial ones. Thus, none of the diagnostics as implemented in Exps. 1–4 replicate Syrett & Koev's finding.

As shown in Fig. 3, Exp. 2 ('asking whether') largely reproduces the pattern from Degen & Tonhauser (2025), namely, that complements of *know* were rated as less at-issue than those of *discover*, followed by *confess*, then *confirm*, and finally *be right*, with the only exception that *confess* and *discover* did not differ in our experiment.

The other three experiments show fewer differences between the mean at-issueness ratings among the clause-embedding predicates. In Exp. 1 (QUD diagnostic), the embedded content of most predicates shows little differentiation, with only two predicates showing a clearly different pattern: the embedded content of *confirm* is rated most at issue, and the embedded content of *be right* is rated less at-issue than the other predicates – the direction of this difference is the opposite to that observed in prior literature and the other experiments. In Exp. 3, no clear differences among clause-embedding predicates are found. Finally, in Exp. 4, the proportions of *yes*-responses are similar for many of the clause-embedding predicates, but the highest proportion is found for content of the complement of *be right*, and it is clearly different from the clause-embedding predicates on the lower end, such as *confirm* and *confess*.

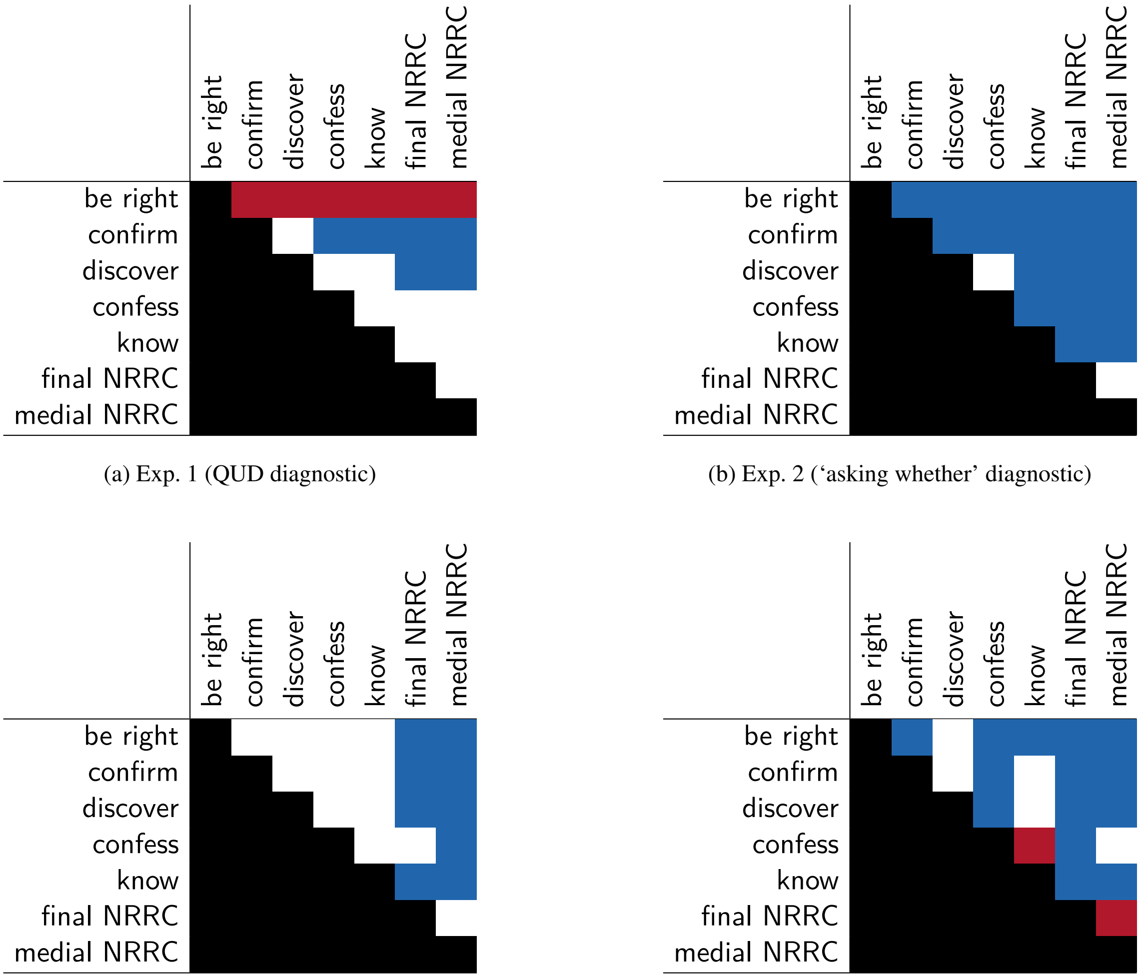
The differences described above are supported by post-hoc pairwise comparisons of the estimated means or proportions for each content, using the `emmeans` package (Lenth 2023) in R (R Core Team 2016), illustrated in Fig. 4. These comparisons were based on mixed-effects beta regression models (Exps. 1–3) and a mixed-effects logistic regression model (Exp. 4), all fit using the `brms` package (Bürkner 2017) using weakly informative priors. The models predicted the ratings[7] from a fixed effect of expression (treatment-coded, with *be right* as reference level), with random intercepts for participants and items. The output of the pairwise comparison were 95% highest density intervals (HDIs) of estimated marginal mean differences between each of the expressions. We assume that two contents differ if their HDI does not include 0.[8]

Accordingly, Figure 4 shows that the only clear positional difference for NRRCs is observed in Exp. 4 ('yes, but' diagnostic), where the sentence-medial ones are rated as more at-issue than sentence-final ones, opposite to the finding in Syrett & Koev (2015). The other three experiments show no differences between sentence-medial and sentence-final NRRCs. The differences between clause-embedding predicates observed in prior work using the 'asking whether' diagnostic are largely replicated in Exp. 2, except for *confess* and *discover*, which do not differ from each other. The other three experiments show fewer differences between clause-embedding predicates, with Exp. 3 showing no differences at all.

We also find that the four experiments differ in the relative ratings assigned to the seven contents, with only limited overlap. Across all experiments, *confirm* and *discover* consistently received higher ratings (at least numerically) than *confess*, which in turn received higher ratings than medial and final NRRCs. For all other pairs of expressions, however, the ordering was not consistent. In particular, the ranking of *be right* relative to most other contents is inconsistent across experiments (e.g., compared to appositive NRRCs): The embedded content of *be right* received the lowest ratings in Exp. 1, but was among the most at-issue in the other three experiments. This difference

[7] To model the ratings in Exps. 1–3 using a beta regression, the ratings were first transformed from the interval [0,1] to the interval (0,1) using the method proposed in Smithson & Verkuilen 2006.

[8] The full model outputs are available in the folder `results+analysis/exps1-4/` in the repository available at https://anonymous.4open.science/r/at-issueness-diagnostics-232C/.



(a) Exp. 1 (QUD diagnostic)

(b) Exp. 2 ('asking whether' diagnostic)

(c) Exp. 3 (direct-dissent diagnostic)

(d) Exp. 4 ('yes, but' diagnostic)

Figure 4: Pairwise differences between expressions, ordered from top to bottom and left to right by increasing mean in Exp. 2 ('asking whether' diagnostic). A white cell means that the 95% HDI of the pair of the row expression and the column expression includes 0, a red cell means that the 95% HDI does not include 0 and that the coefficient is positive (the row expression received a higher rating than the column expression), and a blue cell means that the 95% HDI does not include 0 and the coefficient is negative (the row expression received a lower rating than the column expression).

between the results of the experiments is quantified in the Spearman rank correlations shown in Table 3.[9]

[9] The Spearman rank correlation coefficient, a value between -1 and 1, is a nonparametric measure of rank correlation: the higher the absolute value of the coefficient, the more the relation between the two variables can be described using a monotonic function. If the coefficient is positive, the value of one variable tends to increase with an increase in the other. In the case of our experiments, a coefficient of 1 for two experiments would mean that there is a perfectly monotone increasing relation between the mean ratings of the seven contents in the two experiments: for any two contents c1 and c2, if c1 ranks below c2 in one experiment (that is, the mean rating of c1 is lower than that of c2), then that ranking is preserved in the other experiment.

	Exp. 1	Exp. 2	Exp. 3	Exp. 4
Exp. 1 (QUD diagnostic)		.11	-.29	-.18
Exp. 2 ('asking whether' diagnostic)			.64	.79
Exp. 3 (direct-dissent diagnostic)				.79

Table 3: Spearman rank correlations between the results of Exps. 1–4.

The rank correlations are particularly low for Exp. 1 compared to the other three experiments, at least partly because of the low relative ranking of *be right*. These results suggest that the four diagnostics as implemented in Exps. 1–4 interact differently with the seven contents investigated.

2.3 Discussion

Exps. 1–4 were designed to compare four at-issueness diagnostics that have been widely used in the literature. To assess whether the diagnostics yield consistent findings when applied to the same target contents, we asked whether they reproduce contrasts reported in prior work, particularly (i) the medial versus sentence-final distinction for appositive NRRCs, and (ii) fine-grained differences among the contents of the complements of clause-embedding predicates. The results of Exps. 1–4 suggest that the four diagnostics do differ in whether they reproduce these contrasts, and in the relative order of the seven contents investigated. Three main differences between the four experiments emerge:

i. Although none of the diagnostics as implemented here reproduce the positional effect for appositive NRRCs reported by Syrett & Koev (2015), Exp. 4 (the 'yes, but' diagnostic) is the only diagnostic that finds a difference between medial and final NRRCs. In particular, we found the opposite pattern from Syrett and Koev here: medial NRRCs were judged as more at-issue than final ones. Exps. 1-3 find no significant difference between medial and sentence-final NRRCs.

ii. They differ in the relative order of the seven contents – most notably in the case of the complement of *be right*, which ranks low under the QUD diagnostic but high under the other diagnostics.

iii. The four experiments differ in the extent to which they differentiate between the seven contents investigated, as evidenced by where they find by-content differences: Only the 'asking whether' diagnostic (Exp. 2) comes close to comes close to observing fine-grained differences between the contents of the complements of clause-embedding predicates as in Degen & Tonhauser (2025).

This section discusses the implications of these findings for the interpretation of the diagnostics and for theories of at-issueness.

2.3.1 Positional effects for appositive NRRCs?

One factor that might explain the difference between our findings and those of Syrett & Koev (2015), aside from the stimuli, is the response task. Although both studies used dissent-based diagnostics with target content presented in assertions, Syrett & Koev's 2015 forced-choice implementation differs from our implementation of the dissent-based diagnostics in Exps. 3 and 4.

In Exp. 3 (direct dissent), we elicited naturalness ratings rather than forced-choice judgments. In Exp. 4 ('yes, but'), which did use a forced-choice continuation task, the response options were different from those used by Syrett & Koev. In our implementation (based on Xue & Onea 2011), all options rejected the target content, and participants chose between expressing direct dissent (*no*), or indirect dissent (with *yes, but* or *yes, and*). In contrast, in Syrett & Koev's study, all options expressed direct dissent (with *no*), and participants chose whether to reject the appositive content or the main clause content.

While the findings suggest that positional effects for appositive NRRCs may not be robust across diagnostics, the corresponding task differences suggest that even small changes in the response task used to operationalize a diagnostic, such forced-choice alternatives, might reverse the observed pattern. More generally, different diagnostics vary in whether they detect any positional difference for appositive NRRCs at all. This suggests that previously reported effects may reflect task- or diagnostic-specific effects rather than stable properties of NRRC interpretation. A natural follow-up would be to test whether applying their particular forced-choice schema to our stimuli would reproduce their effect.

2.3.2 Interactions with discourse requirements: the case of *be right*

The diagnostics, as implemented here, yield different rank orderings of contents. The most pronounced ranking difference concerns the complement of *be right*: it receives the lowest ratings in Exp. 1 (QUD diagnostic) and high ratings in the other experiments.

At first glance, this divergence might suggest that the diagnostics track distinct underlying phenomena, but this interpretation is not supported. Both the dual-notion view of at-issueness and the view that distinguishes genuine at-issueness diagnostics from those involving propositional anaphora predict that the QUD diagnostic (Exp. 1) should pattern with the 'asking whether' diagnostic (Exp. 2), and contrast with the dissent-based diagnostics (Exps. 3–4). The observed pattern, in which Exp. 1 diverges from all other diagnostics, is not predicted by these views.

The pattern for *be right* could instead be explained by how the QUD diagnostic interacts with the discourse requirements of *be right*. Specifically, we hypothesize that *be right* places constraints on the prior discourse that conflict with the context introduced by the QUD diagnostic. To see how this explanation might work, first consider thatparticipants gave relatively low naturalness ratings to responses such as (12a), indicating that they did not judge the response as fitting the question.

(12) a. Exp. 1 (QUD diagnostic) with *be right*
Nora: *What did Lucy break?*
Leo: *Danny is right that she broke the plate.*

b. Exp. 2 ('asking whether' diagnostic)
Nora: *Is Danny right that Lucy broke the plate?*

c. Exp. 3 ('direct dissent' diagnostic)
Nora: *Danny is right that Lucy broke the plate.*
Leo: *No, she didn't break the plate.*

d. Exp. 4 ('yes, but' diagnostic)
Nora: *Danny is right that Lucy broke the plate.*
Leo: *Yes, but she didn't break the plate.*
Yes, and she didn't break the plate.
No, she didn't break the plate.

We hypothesize that the low ratings in Exp. 1 arise because *be right* imposes a contextual felicity requirement that a prior commitment by an attitude holder be salient in the discourse. Utterances of the form *X is right that p* require that the common ground or discourse model contains the information that *X* has previously endorsed *p* (or a closely related proposition) (Abusch 2010, p. 50; Anand & Hacquard 2014). In (12a), this requirement is that Danny has previously committed to the proposition that "Lucy broke the plate", as would be the case in the discourse in (13).

(13) a. **Danny:** *Lucy broke the plate.* (QUD: "whether Lucy broke the plate")

b. **Nora:** *What did Lucy break?*
c. **Leo:** *Danny is right that she broke the plate.*

We can understand the ill-formedness of (13) with a QUD-based framework of discourse structure (in the sense of Ginzburg 1996; Roberts 2012; Büring 2003). Within this framework, the contextual requirement of *be right*, that a prior commitment be salient in the discourse, can be understood as a requirement that QUD addressed by the act of making this commitment addresses must be part of the QUD stack. In the case of (13), this means that the question of whether Lucy broke the plate must be part of the QUD stack at the point where *be right* is uttered.

This question is a subquestion of the question in (13b) "What did Lucy break", since every complete answer to the latter entails an answer to the former. The progression from the presupposed subquestion question to the explicitly given superquestion violates Roberts's 2012 constraint on QUD stacks (p. 6:15), which allows only the reverse order (from superquestions to subquestions).[10]

Under the hypothesis put forward here, the low ratings in Exp. 1 do not indicate that the embedded proposition is not at-issue.[11] Rather, they reflect that the response is infelicitous in the given discourse context. In the other diagnostics (12b)–(12d), no conflicting prior discourse is given, allowing the discourse requirements of *be right* to be accommodated without a reduction in acceptability. This highlights that diagnostics may interact with contextual requirements independently of at-issueness, and such interactions must be taken into account when interpreting the results.

2.3.3 Why does the 'asking whether' diagnostic show greater differentiation?

A striking difference between the results of the experiments is that Exp. 2 ('asking whether') showed the greatest differentiation between contents, whereas the other three experiments exhibited a smaller range of means, and fewer statistically supported differences between investigated contents. Among the first 4 experiments, Exp. 2 was the only one that came close to differentiating fine-grained by-content differences as reported in Degen & Tonhauser (2025).

Considering the question whether the existing definitions of at-issueness target the same underlying phenomenon, we might consider whether this empirical difference between diagnostics aligns with the divide emphasized in prior work between diagnostics that have been associated with a QUD-based definition (QUD, 'asking whether') and those that have been argued to target a proposal-based definition (direct-dissent, 'yes, but'). However, this distinction could not explain why the 'asking whether' diagnostic behaves differently from the QUD diagnostic, since both have been linked to the QUD-based definition of at-issueness.

To explain the greater by-content differentiation of the 'asking whether' diagnostic, we therefore focus on properties that set it apart from the other three diagnostics. Two differences are particularly salient: First, it is the only diagnostic where the target content itself occurs in a polar question, whereas in the other diagnostics, the target content appears in a declarative sentence. Second, it is the only diagnostic where participants are asked directly about the intentions of the speaker, while the other ones collect meta-linguistic judgments like acceptability ratings or continuation preferences. Either of these properties might be important for explaining the greater differentiation.

To distinguish between these possibilities, Exps. 5–6., reported in the next section, compare the 'asking whether' diagnostic to a diagnostic that also embeds the target content in a polar question

[10] this constraint can be understood as an informativeness-constraint, which requires that monotone increasing information quantity. in the default case, even strictly increasing quantitiy, since -. where the first two questions engage with the same question, the second one can only be interpreted as an echo question.

[11] When *be right* is excluded, the Spearman rank correlations are:

	Exp. 1	Exp. 2	Exp. 3	Exp. 4
Exp. 1 (QUD diagnostic)		.77	-.09	-.31
Exp. 2 ('asking whether' diagnostic)			.66	.66
Exp. 3 ('direct dissent' diagnostic)				.77

but elicits naturalness ratings to assess at-issueness. Both experiments yielded similarly fine-grained distinctions among contents, supporting the conclusion that the increased differentiation observed in Exp. 2 is driven primarily by interrogative embedding rather than by the response task.

3 Experiments 5 and 6

Exps. 5 and 6 were designed to investigate whether the greater differentiation observed in Exp. 2 ('asking whether') was due to (i) embedding target contents in polar questions or (ii) the response task, testing two hypotheses:

i. **Question-embedding hypothesis:** Differentiation between contents is greater when contents are embedded in a polar question than in a declarative sentence.
ii. **Response-task hypothesis:** Differentiation is greater when participants are asked directly what the utterance is about, compared to tasks such as acceptability judgments or forced-choice continuations.

To test these hypotheses, we compared two diagnostics that both embed the target content in a polar question but differ in the response task. Exp. 5 used the 'asking whether' diagnostic as in Exp. 2 (14a) and Exp. 6 uses a direct-response diagnostic (Amaral et al. 2007; Tonhauser 2012), shown in (14b).

(14) Implementation of the diagnostics in Exps. 5–6

a. Exp. 5 ('asking whether' diagnostic)
Nora: *Is Tom right that Lucy broke the plate?*
Question to participants: Is Nora asking whether Lucy broke the plate?

b. Exp. 6 (direct-response diagnostic)
Nora: *Is Tom right that Lucy broke the plate?*
Leo: *Yes, she didn't break the plate.*
Question to participants: How natural is Leo's response to Nora's question?

This direct-response diagnostic assumes that at-issue content is interpreted as determining the question alternatives (property (ii)). Participants read a dialogue between two named speakers, where the first utters a polar question, which contributes the target content. They then rate the naturalness of a *yes* response that asserts the target content. A response to a question is felicitous only if the target content provides a possible answer, that is, a question alternative (von Stechow 1991). Accordingly, naturalness ratings are expected to be high when the target content is at-issue and low otherwise.

Both experiments measured at-issueness of the contents of the complements of the 20 clause-embedding predicates from Degen & Tonhauser 2025, which were shown in Figure 1.

(15) 20 clause-embedding predicates from Degen & Tonhauser 2025:
acknowledge, admit, announce, be annoyed, be right, confess, confirm, demonstrate, discover, establish, hear, inform, know, pretend, prove, reveal, say, see, suggest, think

Both experiments also assessed the projection of the complement contents. This data was reported in Hofmann et al. 2024. Here we focus on the at-issueness data, which have not yet been reported.

If the greater differentiation between contents found in Exp. 2 (compared to Exps. 1, 3, and 4) was driven primarily by presenting the target content in a polar question, we expect Exps. 5 and 6 to show similar fine-grained by-content differentiation of at-issueness. In contrast, if Exp. 5 again shows greater differentiation than Exp. 6, this would support the response-task hypothesis, indicating that the task, rather the question embedding, drives the effect.

3.1 Methods

3.1.1 Participants

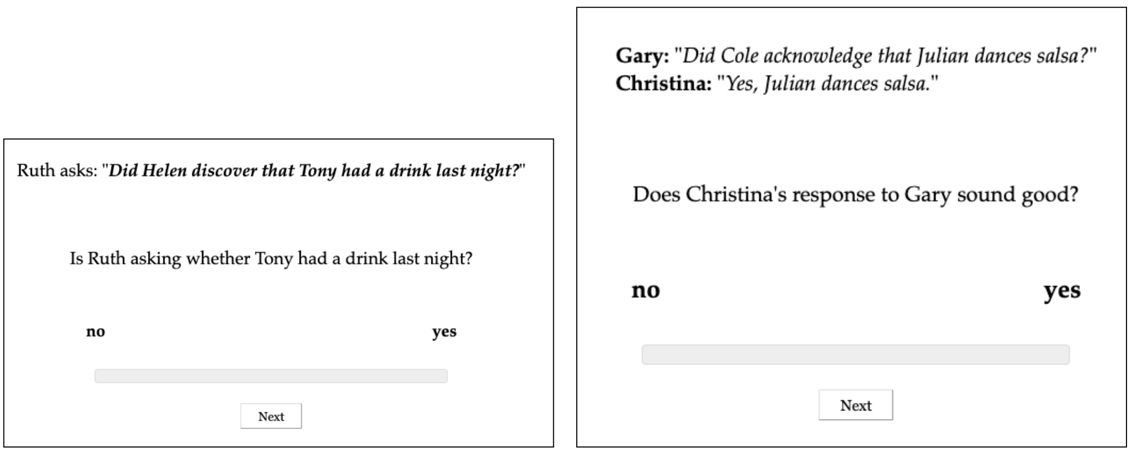
We recruited 284 participants on Amazon's Mechanical Turk platform for Exp. 5 and 250 participants on Prolific for Exp. 6.[12] Participants recruited on Mechanical Turk had U.S. IP addresses and at least 99% of previous HITs approved. Participants recruited on Prolific had registered as U.S.-born native speakers of English residing in the USA and had an approval rate of at least 99%. Table 4 summarizes the age and gender distributions of the recruited participants.

	recruited	ages (mean age)	f/m/nb/dnd
Exp. 5 (asking whether)	284	19-74 (38.2)	–/–/–/–
Exp. 6 (direct response)	250	18-58 (25.5)	201/43/6/0

Table 4: Information about the participants recruited in Exps. 5-6 (f = female, m = male, nb = nonbinary, dnd = did not disclose; gender information was not collected in Exp. 5).

3.1.2 Materials and procedure

As shown in Fig. 5, the response task differed between experiments. In Exp. 5 ('asking whether' diagnostic, panel (a)), participants judged whether the question was about the content of the complement clause, using a slider marked 'no' on one end (coded as 0) and 'yes' on the other (coded as 1). In Exp. 6 (direct-response diagnostic, panel (b)), participants rated whether the response to the first speaker sounds good, on a slider marked 'no' (coded as 0) on one end and 'yes' on the other (coded as 1). In the initial instructions, participants were asked to imagine that they are at a party and that, when walking into the kitchen, they overhear somebody say something to somebody else. Across both experiments, the responses were coded so that 1 meant that the content of the embedded clause was rated as at-issue and 0 as not-at-issue.



(a) Exp. 5: 'asking whether' diagnostic

(b) Exp. 6: direct response diagnostic

Figure 5: Sample trials in (a) Exp. 5 and (b) Exp. 6.

In both experiments, target stimuli were created by combining each of the 20 clause-embedding predicates in (15) with one of 20 items (listed in Supplement B). The same six control stimuli were

[12] Exp. 5 was run in August 2019 and Exp. 6 in August 2021.

included in both experiments. The contents targeted in these controls (see Supplement C) were expected to be at-issue and served as attention checks. For each participant, the 20 predicates were randomly paired with 20 unique items, yielding one target stimulus per predicate. Participants thus saw 26 stimuli in total: 20 target items and 6 controls.[13] Trial order was randomized for each participant. After completing the experiment, participants filled out a short optional demographic survey. To encourage truthful responses, participants were told that they would be paid no matter what answers they gave in the survey.

3.1.3 Data exclusion

We excluded the data of participants who did not self-identify as native speakers of American English, and of participants whose responses to one of the two control trials was more than 2 sd away from the group mean. Table 5 shows how many participants were excluded in each experiment, the properties of the remaining participants, and the number of data points that entered into the analyses.

	exclusion criterion		**remaining participants**			data
	language	controls	ages (mean age)	f/m/nb/dnd	total	points
Exp. 5 (asking whether)	7	35	21-74 (39.2)	–/–/–/–	242	6292
Exp. 6 (direct response)	5	24	18-58 (24.9)	187/29/5/0	220	5746

Table 5: Information from Exps. 5-6 about the number of participants whose data was excluded based on their self-declared language and language variety ('language'), the controls, and the variance of their responses, about the remaining participants, and about the number of at-issueness data points that entered into the analysis.

3.2 Results and discussion

Fig. 6 shows the results of the two experiments by embedding predicate: panel (a) plots mean 'asking whether' ratings in Exp. 5, and panel (b) plots mean naturalness ratings in Exp. 6 (direct-response).

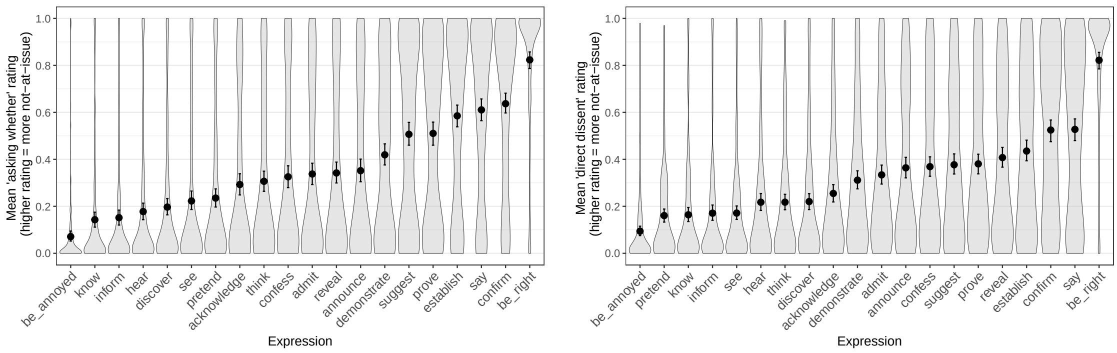
Overall, the results of two experiments yield similar patterns. The ranges of by-content means are comparable (Exp. 5 range: .75, .07 to .82; Exp. 6 range: .73, .09 to .82), and the relative orderings of contents by their means align closely, as reflected in a strong Spearman rank correlation ($r_s = .93$).

Crucially, both experiments show fine-grained by-content differentiation. For instance, for the subset of predicates also tested in Exps. 1–4, Exps. 5 and 6 reproduce the ordering expected based on Degen & Tonhauser (2025) and show pairwise differences in the predicted directions for adjacent contents, with differences between *know, discover, confess, confirm, be right* with increasing mean at-issueness ratings in that order. Overall, the results in Exps. 5 and 6 are similar in revealing fine-grained differences among many rank-adjacent predicates.

There are also some differences between the experiments, concerning a small number of local reversals in the middle of the scale. For example, the content of the complement of *demonstrate* comes out as more at-issue than that of *announce*, *confess*, and *reveal* under the 'asking whether' diagnostic as implemented in Exp. 5, but as less at-issue than the content embedded under these predicates under the implementation of the direct-response diagnostic in Exp. 6.

Fig. 7 presents the results of post-hoc pairwise comparisons of the estimated means for each content. As in Section 2, we used `emmeans` (Lenth 2023) in R (R Core Team 2016) applied to

[13] Each participant saw their set of 26 stimuli twice, once in the projection block and once in the at-issueness block. Block order was randomized. As mentioned above, we focus here on the at-issueness data.



(a) Exp. 5 ('asking whether' diagnostic) (b) Exp. 6 ('direct assent' diagnostic)

Figure 6: Results of Exps. 5–6. The panels show the mean ratings by expression for (a) Exp. 5 (asking whether diagnostic) and (b) Exp. 2 (direct assent diagnostic). Error bars indicate 95% bootstrapped confidence intervals. Violin plots show the kernel probability density of individual participants' ratings.

mixed-effects beta regression models fit with `brms` (Bürkner 2017) using weakly informative priors. The models predicted ratings[14] from a fixed effect of content (treatment-coded, with *be right* as the reference level). The pairwise comparisons support the above generalizations: while the two diagnostics as implemented here differ in which specific contrasts they detect, their overall patterns are highly consistent. Both diagnostics reveal fine-grained at-issueness differences between contents. At the same time, there are some differences, namely some contrasts that reached significance in experiment did not in the other.

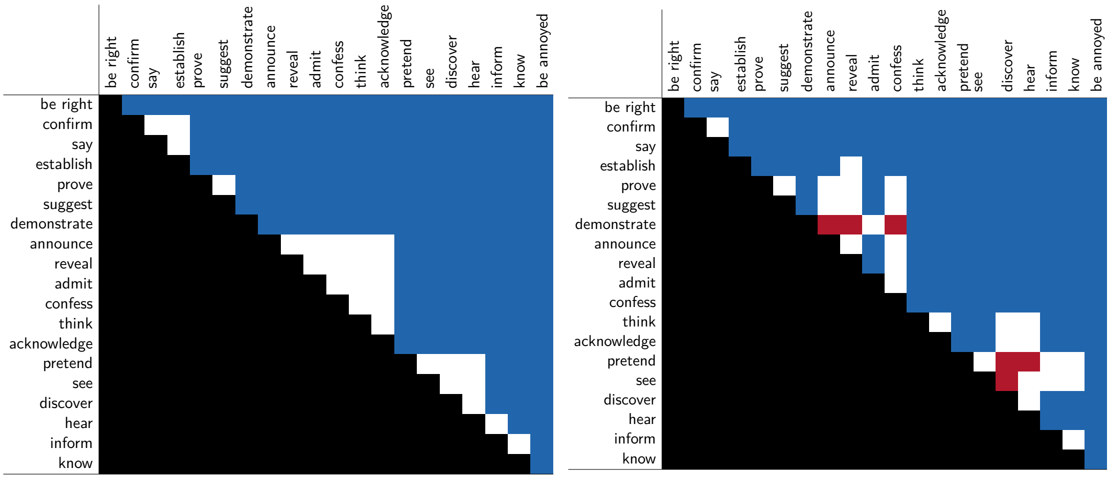
Taken together, Exps. 5 and 6 suggest that the two diagnostics as implemented here ('asking whether' and direct-response) yielded highly similar results for the 20 contents investigated, including fine-grained by-predicate differentiation. Although they differ in response task, both present the target content in a question and both yield strong by-content differentiation. This pattern supports the question-embedding hypothesis over the response-task hypothesis.

Under the response-task hypothesis, differentiation should be greater when participants are asked directly what an utterance is about than in tasks such as acceptability judgments or forced-choice continuations, and the direct-response diagnostic (Exp. 6) should show lower differentiation than the asking-whether diagnostic (Exps. 2 and 5). This prediction is not reflected in the data.

Instead, both Exps. 5 and 6 exhibit fine-grained pairwise differences across contents, as predicted by the question-embedding hypothesis. This hypothesis attributes increased differentiation to presenting the target content in a polar question, a property shared by Exps. 2, 5, and 6 but absent in Exps. 1, 3, and 4, where the target content was presented in a declarative.

Nonetheless, small differences remain, for example in which by-content contrasts reached significance and in the precise rank ordering of contents. Since the two experiments used identical embedding contexts for the target content and both tested property (ii), which has been linked to a QUD-based definition of at-issueness, these differences may stem from the response tasks used: judging what a question is about (Exp. 5) versus evaluating the naturalness of a response (Exp. 6). This suggests that response task differences can affect at-issueness judgments in subtle ways.

[14] To model the ratings using a beta regression, the ratings were first transformed from the interval [0,1] to the interval (0,1) using the method proposed in Smithson & Verkuilen 2006.



(a) Exp. 5 ('asking whether' diagnostic)

(b) Exp. 6 (direct-response diagnostic)

Figure 7: Pairwise differences between expressions, ordered from by increasing mean in Exp. 5 ('asking whether'). White cells indicate that the 95% HDI of the difference includes 0. Red cells indicate a positive difference (the row expression received a higher rating than the column expression), and blue cells indicate a negative difference.

4 General discussion

This paper reported six experiments addressing two overarching questions: (i) whether different diagnostics yield consistent results when applied to the same contents, and (ii) whether differences between diagnostics reflect distinct underlying notions of at-issueness.

The experiments provide a clear answer to the first question: the diagnostics did not yield consistent results. Across Exps. 1–4, they differed in the extent to which they replicated previously reported differences among contents. A key methodological implication is therefore that at-issueness diagnostics cannot be treated as interchangeable. Theoretical claims obtained using one diagnostic should be understood relativize to that diagnostic, and results obtained with one task should not be assumed to automatically generalize to others.

To begin addressing the second question, we asked what drives the observed diagnostic differences. Taken together, the experiments provide converging evidence that diagnostics that present target contents in polar questions yield greater by-content differentiation than diagnostics that present the same contents in declaratives. When target contents were presented in a polar question, as in the 'asking whether' diagnostic (Exps. 2, 5) and the direct-response diagnostic (Exp. 6), we observed high by-content differentiation, reproducing fine-grained by-content differences reported in prior work. In contrast, when the same contents were presented in declarative sentences, as in the QUD diagnostic (Exp. 1) and the dissent-based diagnostics (Exps. 3–4), the resulting ratings showed substantially less differentiation. This suggests that clause type and the illocutionary force of the utterance in which the target content is presented need to be taken into consideration, when interpreting outcomes of at-issueness diagnostics.

At the same time, not all differences between diagnostics are explained by this contrast. We also observed diagnostic-specific effects tied to particular contents and task designs. For example, we found that diagnostics differ in whether they detect positional effects for appositive NRRCs, and even in the direction of these effects (Section 2.3.1). Another example is the complement of *be right*, which ranked low under the QUD diagnostic (Exp. 1) despite ranking high in the other experiments (Section 2.3.2).

These findings bear on the second research question. Empirical divergence between different diagnostics raises the possibility that they track distinct underlying discourse phenomena, such as QUD-based at-issueness, proposal-based at-issueness, or anaphoric availability. However, such an interpretation is only supported if the observed differences align systematically with the contrasts predicted by those theoretical views. Our results do not show this an alignment. The clearest diagnostic contrast we found, namely the difference between polar-question-based and assertion-based diagnostics, cuts across existing theoretical classifications. Other differences are more plausibly attributed to contextual requirements of particular expressions, or response-task differences. This does not rule out the possibility that some diagnostics are sensitive to distinct underlying notions, but it does suggest that much of the observed variation arises from factors other than the theoretical distinctions currently discussed in the literature.

The remainder of this section develops these points in more detail. Section 4.1 discusses the methodological implications of diagnostic non-convergence. Section 4.2 considers the role of question embedding and speaker commitment in producing greater by-content differentiation. Section 4.3 then returns to the broader question of whether the observed diagnostic differences support distinct underlying notions of at-issueness.

4.1 Methodological implications

A methodological implication of these diagnostic differences is that they cannot be treated as interchangeable. Theoretical claims based on one diagnostic should therefore be relativized to the diagnostic used (see also Snider 2017b; a; 2018; Koev 2018; Korotkova 2020). Therefore, empirical investigations should carefully consider that results may depend on whether the target content was presented in an assertion or polar question, contextual requirements of the expressions used, or the particular task design.

JT: why is the methodological implication not a separate subsection?

LH: Okay, I am leaving this here in a subsection for now. Maybe you would like to add something, or do you think that this merits its own subsection as it is? Alternatively, it could be used above, before the paragraph that starts "The remainder of this section…"?

4.2 Speech act, at-issueness, and speaker commitments

A central empirical generalization emerging from our experiments is that diagnostics differ systematically in the degree of by-content differentiation they yield depending on how the target content is embedded. Diagnostics that present target contents in declarative sentences show reduced differentiation, whereas diagnostics that present the same contents in polar questions show greater differentiation. This raises the question of why the embedding context should have such a strong and systematic effect on diagnostic outcomes.

We hypothesize that this pattern may reflect a ceiling effect in assertion-based diagnostics, driven by how different speech acts relate propositional content to speaker commitment. We assume that the declaratives in these diagnostics are interpreted as assertions, which involve speaker commitment to the truth of the at-issue content, whereas the polar questions are interpreted as information-seeking questions, which do not involve such commitment. On this view, the illocutionary force of the embedding utterance determines how the at-issue content relates to the speaker's commitments, and thereby affects how easily it can be distinguished from not-at-issue content.

The key contrast concerns the epistemic status of at-issue and not-at-issue content in polar questions versus assertions. In a polar question like (16), the at-issue content determines the set of alternatives under consideration (16a), but the speaker is not committed to any of these alternatives. In contrast, content that is (more) not-at-issue, like the NRRC content in (16b), is more likely to project, that is treated as true throughout the context set, and is therefore part of the speaker's commitments (Abusch 2010; Simons et al. 2010; Abrusán 2011; Tonhauser et al. 2018; Degen & Tonhauser 2025).

(16) *Is Greg, who bought a car, envied by his neighbor?*

a. At-issue contribution: determining the question alternatives
{Greg, who bought a car, is envied by his neighbor,
Greg, who bought a car, is not envied by his neighbor}

b. Not-at-issue contribution: *Greg bought a car.*

As a result, the at-issue and not-at-issue content of questions differ sharply in their *epistemic status* relative to the speaker commitments: their not-at-issue content is associated with speaker commitment, while their at-issue content is not.

In assertions, this contrast is weakened. An assertion of a declarative sentence like (17) both proposes to add the content of the at-issue main clause (17a) to the common ground and commits the speaker to its truth (Stalnaker 1978; Gunlogson 2001). In addition, the not-at-issue content (17b) is also treated as part of the speaker's commitments.

(17) *Greg, who bought a car, is envied by his neighbor.*

a. *Greg is envied by his neighbor.* (at-issue main clause content)

b. *Greg bought a car.* (not-at-issue content)

In (17), both the at-issue and not-at-issue content are associated with speaker commitment. The epistemic distinction between the two types of content is therefore reduced.

We propose that this difference in epistemic contrast could affect how participants identify at-issue content in experimental tasks. Diagnostics of at-issueness require participants to evaluate which content an utterance is primarily about. In assertions, the at-issue main clause content is the content to which the speaker is committed. Participants may therefore use speaker commitment as a cue for identifying the main point of the utterance. However, not-at-issue content, such as appositive material, is also typically treated as part of the speaker's commitments, due to projection. As a result, both at-issue and not-at-issue content satisfy the same criterion: they are contents the speaker is committed to. We suggest that this similarity makes the two types of content harder to distinguish in assertion-based diagnostics. This would lead to not-at-issue content being rated more similarly to at-issue content, giving rise to overall higher ratings for not-at-issue content and reduced differentiation across contents, essentially creating a ceiling effect.

This reduction in contrast does not arise in polar questions: Although the not-at-issue content of polar questions is also associated with speaker commitment due to projection, their at-issue content is not. As a result, speaker commitment cannot be used as a cue for identifying the main point of the utterance in questions, and interpreters must rely on other aspects of the utterance instead. Here, not-at-issue content is not rated more similarly to at-issue content, and the epistemic contrast between the two types of content is preserved, allowing for greater differentiation. On this view, polar questions do not so much enhance differences as leave them visible, whereas assertions tend to mask them.

Importantly, this hypothesis does not depend on the target content being clearly not-at-issue, as in the case of appositive NRRCs, but it predicts that assertion-based diagnostics can obscure more subtle differences in at-issueness as well. For contents whose status is less clearly determined by conventional meaning, such as complements of verbs like *discover* or *reveal*, reduced epistemic contrast in assertions may compress differences between at-issue and not-at-issue content as well. In the context of a diagnostic, the utterance of a declarative sentence (18) signals that speaker is committed to the main clause at-issue content (18b), and we would be diagnosing the at-issueness of the target content (18a).

(18) *Helen confirmed that Tony had a drink last night.*
  a. p : *Tony had a drink last night* (target content)
  b. q : *Helen confirmed* p (main-clause content)

Now, if (18a) (p) is interpreted as more at-issue, then at-issueness ratings will be higher. If p is interpreted as less at-issue, at-issueness ratings should in principle be lower, but here, ratings for less at-issue content could be inflated due to projection and speaker commitment. If p is (more) not-at-issue, it is also more likely to project, leading to speaker commitment (Tonhauser et al. 2018; Degen & Tonhauser 2025). Under a gradient view of at-issueness, we could, for instance, articulate this in terms of the gradient projection principle (Tonhauser et al. 2018), which suggests that the more at-issue it is, the more projective it also is, and thus, the more associated with speaker commitment. In turn, the more a content is associated with speaker commitment, the harder it is to distinguish from the at-issue main clause in assertions.

The hypothesis laid out here can explain the main pattern observed in our experiments. Diagnostics that embed target contents in polar questions, such as the 'asking whether' diagnostic (Exps. 2, 5) and the direct-response diagnostic (Exp. 6), preserve the epistemic contrast between at-issue and not-at-issue content and accordingly yield high by-content differentiation. Diagnostics that embed target contents in assertions, on the other hand, such as the QUD diagnostic (Exp. 1) and the dissent-based diagnostics (Exps. 3–4), compress differences between at-issue and not-at-issue-content, and lower overall differentiation.

In sum, the difference in by-content differentiation between question-based and assertion-based diagnostics can be understood as a consequence of how illocutionary force determines the relationship between propositional contents and speaker commitment. In polar questions, there is a clear distinction between the epistemic status of at-issue and not-at-issue content, allowing for discrimination, whereas assertions reduce this distinction, leading to a compression of judgments.

To our knowledge, no previous literature has proposed that the speech act in which target content is presented can itself be a major driver (or inhibitor) of differentiation. Here, we saw that polar questions more transparently reveal how different contents can be interpreted as at-issue or not-at-issue, and suggested that this may be because the two contents differ in their epistemic status. Assertions, in contrast, commit the speaker both to the at-issue assertion and the not-at-issue content, thereby obscuring the distinction. Consequently, diagnostics that embed the target content in questions provide a particularly sensitive test of at-issueness. Embedding target content in questions therefore allows for detection of more fine-grained differences in at-issueness associated with the conventional meaning of particular expressions, whereas assertions introduce a confounding factor: the speaker commits to both types of content, reducing differentiation in at-issueness judgments.

4.3 Diagnostic differences and notions of at-issueness

Because the diagnostics are grounded in distinct assumptions about what it means for content to be at-issue, the observed differences between diagnostics also bear on the broader question of whether existing diagnostics track a common underlying notion of at-issueness or operationalize distinct phenomena. This subsection considers how the differences between diagnostics we observed relate to patterns that would be expected based on views that assume that different diagnostics target a unified phenomenon of at-issueness (e.g., Tonhauser 2012; AnderBois et al. 2015; Faller 2019), different underlying notions (Koev 2018), or that some diagnostics target at-issueness, while others diagnose the availability of the target content for propositional anaphora (Snider 2017a; a; 2018; Korotkova 2020).

Recall the definitions of QUD at-issueness and proposal at-issueness, repeated here:

(1) QUD at-issueness: (based on Simons et al. 2010, p. 323)
A content *m* is at-issue in a context *c* iff
a. *m* is relevant[15] to the QUD in *c*, and
b. the speaker has a recognizable intention to address the QUD via *m*.

(2) Proposal at-issueness: (Koev 2013, pp. 50–51)
A proposition *p* is at-issue in a context *c* iff
a. *p* is a proposal in *c* and
b. *p* has not been accepted or rejected in *c*.

These definitions differ in their theoretical grounding, and the literature differs in how this difference should be interpreted. One possibility is that they characterize different perspectives on a single underlying phenomenon (e.g., Tonhauser 2012; AnderBois et al. 2015; Faller 2019). On this view, diagnostics may differ in implementation, but they should ultimately pattern together to the extent that they probe the same discourse property. Another possibility is that the two definitions characterize two different underlying phenomena (Koev 2018). On this view, at least some diagnostic differences should reflect this theoretical divide. A third possibility is that some apparent diagnostic differences do not track different notions of at-issueness at all, but instead arise because certain diagnostics are sensitive to other properties, such as propositional anaphoric availability (Snider 2017b; a; 2018; Korotkova 2020).

These positions lead to different expectations about what kind of empirical divergence would count as evidence for distinct underlying notions. The dual-notion view predicts that at least some diagnostic differences reflect a split between diagnostics aligning with the proposal-based definition, i.e., the dissent-based diagnostics (Exps. 3–4) and those aligning with the QUD-based definition (i.e, the QUD diagnostic from Exp. 1; the 'asking whether' diagnostic from Exp. 2; and the direct response diagnostic from Exp. 6). The anaphoric-accessibility view would expect systematic differences between diagnostics that involve propositional anaphora, like the dissent-based diagnostics (Exps. 3–4) and the direct-response diagnostic (Exp. 5), and those that do not, like the QUD diagnostic (Exp. 1) and the 'asking whether' diagnostic (Exp. 2). The unified view would be supported by convergent results across diagnostics, but it is also consistent with diagnostic differences attributed to factors that are orthogonal to differences in at-issueness itself.

In the following, we discuss whether our results provide support for any of these views. We highlight three main points.

First, the contrast that aligns with the clearest difference between diagnostics we found – whether target contents are presented in polar questions versus assertions – may appear related to the distinction between QUD-based and proposal-based definitions of at-issueness, since both involve a question versus assertion contrast. However, as discussed in Section 2.3.3, this pattern is not predicted by the differences in the underlying notions targeted by the diagnostics that have been proposed in the literature. The dual-notion view could only provide an explanation for differences between diagnostics targeting property (i) (the QUD diagnostic) and those targeting property (iii) (direct dissent and 'yes, but'). The anaphoric-availability view would be supported by differences between diagnostics targeting properties (i) or (ii)[16] such as the QUD diagnostic and 'asking whether' and those targeting property (iii). Our main contrast, however, cuts across these groupings. Although the main diagnostic difference is not straightforwardly predicted by either the dual-notion view or the anaphoric-availability view, it remains in principle compatible with all three positions.

[16] The status of the direct-response diagnostic further complicates the picture. It targets property (ii), which Snider aligns with QUD-based at-issueness, but it also involves the response particle yes, which has been argued to diagnose anaphoric availability in property (iii) diagnostics. It is therefore unclear how this diagnostic should be classified under the anaphoric-availability view.

Second, the diagnostics differ in how they interact with contextual requirements imposed by particular expressions. A clear example is the behavior of *be right* under the QUD diagnostic, where low QUD-match ratings do not plausibly reflect not-at-issueness, but instead arise from a pragmatic incompatibility: *be right* presupposes a discourse structure that conflicts with the assumptions made by the diagnostic (see Section 2.3.2). A related point is raised by Snider (2017a; 2018) who emphasized that the direct-dissent and 'yes but' diagnostic involve propositional anaphora in the interpretation in the response particles *yes/no*. Different diagnostics impose different requirements on discourse structure, common ground, and anaphoric accessibility, all of which can independently affect acceptability judgments. This supports the conclusion that some differences between diagnostics arise from interactions with expression-specific contextual requirements rather than from differences in at-issueness per se.

Third, some small differences between diagnostics can be attributed to response-task effects. This is evident in the subtle differences between Exps. 5 and 6 (see Section 3.2), which differed only in response format, and in our failure to replicate Syrett & Koev's 2015 reported positional effect for appositive NRRCs (see section 2.3.1). While they found sentence-final appositives to be more at-issue than sentence-medial ones under a version of the direct-dissent diagnostic, our assent/dissent-based *yes, but* diagnostic in Exp. 4 showed a difference in the opposite direction, while the other experiments found no difference between them (Exp.1–3). Taken together, these results provide no evidence for a stable notion of at-issueness under which sentence-final appositives are systematically more at-issue than medial ones (or the other way round), and they highlight how even small task differences can affect outcomes.

At the same time, among most pairs from Exps. 1–4, the Spearman rank correlation was at least moderate (when excluding *be right*, see footnote 11), suggesting some convergence among diagnostics, and that the diagnostics, as implemented there, are at least partially sensitive to a shared underlying property.

Our data does not provide conclusive evidence about whether or not the diagnostics track fundamentally different underlying notions of at-issueness. However, the fact that many of the differences find explanations outside this question, seems more compatible with assuming that the diagnostics offer different, partially overlapping windows onto a single discourse phenomenon, that interacts with multiple discourse factors (speaker commitments, speech acts, contextual requirements, and the given alternatives) in complex and sometimes subtle ways, so that even small differences in task design can make a difference.

5 Conclusion

This paper presented a systematic experimental comparison of at-issuness diagnostics, motivated by two questions about experimental diagnostics of at-issueness: whether different diagnostics yield consistent results when applied to the same contents, and whether any differences between diagnostics support the view that they are sensitive to distinct underlying phenomena.

Across six experiments, we tested five diagnostics: the QUD diagnostic, the 'asking whether' diagnostic, the direct-dissent diagnostic, the 'yes, but' diagnostic, and the direct-response diagnostic. Our results provide experimental confirmation for claims from the theoretical literature that there are empirical differences between at-issueness diagnostics (Snider 2017b; a; 2018; Koev 2018; Faller 2019; Korotkova 2020). Since the diagnostics did not yield fully consistent results, they cannot be treated as interchangeable.

The experiments also revealed a robust source of variation: diagnostics that presented target contents in polar questions showed greater by-content differentiation than diagnostics that presented the same contents in assertions. We argued that this pattern shows that illocutionary force and speaker commitments affect the outcomes of at-issueness diagnostics, and that these factors should be taken into account when comparing diagnostic results.

Our findings do not provide evidence that different diagnostics target distinct underlying phenomena. Although the diagnostics differed empirically, the clearest differences did not align in any straightforward way with existing proposals distinguishing QUD-based at-issueness, proposal-based at-issueness, and anaphoric availability. Because many of the observed differences can be attributed to factors other than the theoretical distinctions currently discussed in the literature, our results are more compatible with the view that the diagnostics offer different, partially overlapping windows onto a single discourse phenomenon that interacts with multiple discourse factors in complex and sometimes subtle ways (e.g., speaker commitments, illocutionary force, contextual felicity requirements, and the given alternatives in forced-choice tasks).

However, because our results do not rule out the possibility that some diagnostics are sensitive to distinct underlying notions, it remains an important open question for future work what the results of different diagnostics can reveal about whether they reflect a shared underlying notion of at-issueness.

Data accessibility statement

The experiments, data and R code for generating the figures and analyses of the experiments reported in this paper are available at https://anonymous.4open.science/r/at-issueness-diagnostics-232C/.

Ethics and consent

All experiments were conducted with approval from the ethics review committee of [university name redacted for review].

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Supplements

A Control stimuli in Exps. 1–4

The examples in (1)-(4) provide the two control stimuli used in each of Exps. 1-4. For the a.-examples, participants were expected to give a 'totally fits' response (Exp. 1), a 'yes' response (Exp. 2), a 'totally natural' response (Exp. 3), and a 'no' response (Exp. 4); for the b.-examples, the opposite response was expected. The numbers after each example identify the mean ratings (Exps. 1-3) or the proportion of 'no' responses (Exp. 4) after excluding participants who did not self-identify as native speakers of American English (but before excluding participants on the basis of these controls), showing that the control stimuli worked as intended.

(1) Control stimuli in Exp. 1 (QUD diagnostic)

a. Mary: Which course did Ava take?
John: She took the French course. (.97)

b. Jennifer: What does Betsy have?
Robert: She loves dancing salsa. (.07)

(2) Control stimuli in Exp. 2 ('asking whether' diagnostic)

a. Mary: Did Arthur take a French course?
Question to participants: Is Mary asking whether Arthur took a French course? (.96)

b. Robert: Does Betsy have a cat?
Question to participants: Is Robert asking whether Betsy loves apples? (.02)

(3) Control stimuli in Exp. 3 ('direct dissent' diagnostic)

a. Mary: Arthur took a French course.
Lily: No, he took a Spanish course. (.87)

b. Robert: Betsy has a cat.
Maximilian: No, she doesn't like apples. (.05)

(4) Control stimuli in Exp. 4 ('yes, but' diagnostic)

a. Mary: Arthur took a French course.
Lily: Yes, but Lisa loves cats. / Yes, and he didn't take a French course. / No, he didn't take a French course. (.95)

b. Robert: Betsy has a cat.
Maximilian: Yes, but she is good at math. / Yes, and she loves it so much. / No, she doesn't like apples. (0)

B 20 clauses

The contents of the following 20 clauses, which realized the complements of the 20 clause-embedding predicates, were investigated in Exps. 5–6:

i. Mary is pregnant.
ii. Josie went on vacation to France.
iii. Emma studied on Saturday morning.
iv. Olivia sleeps until noon.
v. Sophia got a tattoo.
vi. Mia drank 2 cocktails last night.
vii. Isabella ate a steak on Sunday.
viii. Emily bought a car yesterday.
ix. Grace visited her sister.
x. Zoe calculated the tip.
xi. Danny ate the last cupcake.
xii. Frank got a cat.
xiii. Jackson ran 10 miles.
xiv. Jayden rented a car.
xv. Tony had a drink last night.
xvi. Josh learned to ride a bike yesterday.

xvii. Owen shoveled snow last winter.
xviii. Julian dances salsa.
xix. Jon walks to work.
xx. Charley speaks Spanish.

C Control stimuli in Exps. 5–6

The control stimuli in Exps. 5–6 were the contents of the main clause polar questions in (5). The non-restrictive relative clauses (NRRCs), given in parentheses in (5), were included in Exp. 6, where at-issueness was measured with an assent diagnostic. The control stimuli here consisted of two clauses (like the target stimuli), to allow the relevant speaker to assent with one of two clauses.

(5) Sentences for control stimuli in in question embedding experiments (Exps. 5–6)
   a. Do these muffins (, which are really delicious,) have blueberries in them?
   b. Does this pizza (, which I just made from scratch,) have mushrooms on it?
   c. Was Jack (, who is my long-time neighbor,) playing outside with the kids?
   d. Does Ann (, who is a local performer,) dance ballet?
   e. Were John's kids (, who are very well-behaved,) in the garage?
   f. Does Samantha (, who is really into fashion,) have a new hat?

We expected participants to give low responses on the at-issueness diagnostics for the control stimuli in (5), indicating that the main clause content is at-issue.